

Research Assistant Agent: A Survey

Abstract

Large language models have enabled autonomous research assistants that can plan, retrieve and synthesize scientific literature. This paper surveys the core components: retrieval-augmented generation, tool use, and multi-agent coordination.

1. Retrieval

Retrieval-augmented generation (RAG) couples a vector index with a generative model. Hybrid retrieval combines dense and sparse signals via reciprocal rank fusion to improve recall on scientific corpora.

2. Multi-Agent Coordination

A supervisor delegates sub-tasks to specialist agents. A shared blackboard lets experts exchange intermediate results, while chain passing forwards a primary product from one agent to the next.

3. Long-term Memory

Conversation memory is persisted to SQLite so that findings survive restarts and can be recalled across sessions with full-text search.